National prevalence of vision impairment and blindness and associated risk factors in adults aged 40 years and older with known or undiagnosed diabetes: results from the SMARTIndia cross-sectional study

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- Centre for Intelligent Healthcare

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- Overview
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Abstract

Background: National estimates of the prevalence of vision impairment and blindness in people with diabetes are required to inform resource allocation. People with diabetes are more susceptible to conditions such as diabetic retinopathy that can impair vision; however, these are often missed in national studies. This study aims to determine the prevalence and risk factors of vision impairment and blindness in people with diabetes in India.

Methods: Data from the SMART-India study, a cross-sectional survey with national coverage of 42147 Indian adults aged 40 years and older, collected using a complex sampling design, were used to obtain nationally representative estimates for the prevalence of vision impairment and blindness in people with diabetes in India. Vulnerable adults (primarily those who did not have capacity to provide consent); pregnant and breastfeeding women; anyone deemed too ill to be screened; those who did not provide consent; and people with type 1 diabetes, gestational diabetes, or secondary diabetes were excluded from the study. Vision impairment was defined as presenting visual acuity of 0·4 logMAR or higher and blindness as presenting a visual acuity of 1·0 logMAR or higher in the better-seeing eye. Demographic, anthropometric, and laboratory data along with geographic distribution were analysed in all participants with available data. Non-mydriatic retinal images were used to grade diabetic retinopathy, and risk factors were also assessed.

Findings: A total of 7910 people with diabetes were included in the analysis, of whom 5689 had known diabetes and 2221 were undiagnosed. 4387 (55·5%) of 7909 participants with available sex data were female and 3522 (44·5%) participants were male. The estimated national prevalence of vision impairment was 21·1% (95% CI 15·7–27·7) and blindness 2·4% (1·7–3·4). A higher prevalence of any vision impairment (29·2% vs 19·6%; p=0·016) and blindness (6·7% vs 1·6%; p<0·0001) was observed in those with ungradable images. In known diabetes, diabetic retinopathy (adjusted odds ratio [aOR] 3·06 [95% CI 1·25–7·51]), vision-threatening diabetic retinopathy (aOR 7·21 [3·52–14·75]), and diabetic macular oedema (aOR 5·41 [2·20–13·33]) were associated with blindness in adjusted analysis. Common sociodemographic risk factors for vision impairment and blindness include older age, lower educational attainment,
and unemployment.

Interpretation: Based on the estimated 101 million people with diabetes in 2021 and the interpretation of the data from this study, approximately 21 million people with diabetes have vision impairment in India, of whom 2·4 million are blind. Higher prevalence is observed in those from lower socio-economic strata and policy makers should focus on these groups to reduce inequalities in health care.

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- **Prevalence**
  Medicine and Dentistry
  100%
- **Diabetes**
  Medicine and Dentistry
  100%
- **Visual Impairment**
  Medicine and Dentistry
  100%
- **Cross Sectional Study**
  Medicine and Dentistry
  100%
- **Amaurosis**
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- **Diabetic Retinopathy**
  Medicine and Dentistry
  40%

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Odds Ratio
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30%

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